

The growth-law correction constant $\delta = \log R_\infty$ of the quadratic polynomial continued fraction: structure, a closed-form bracket, and certified high-precision values

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Abstract

For integers $A, C \geq 1$ and $B \geq 0$ let $b_k = Ak^2 + Bk + C$ and let q_n be the standard continuant denominators of the polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$ ($q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$). A companion note proves the growth law $\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \kappa/n + O(1/n^2)$, in which the product constant $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ is closed-form but the correction $\delta = \log R_\infty > 0$, with $R_\infty = \lim_n q_n / \prod_{k \leq n} b_k$, is left undetermined. This note *characterizes* δ . We show that the scaled limit R_∞ is exactly the *independence polynomial of a path graph* evaluated at the weights $u_n = 1/(b_{n-1}b_n)$, giving a cluster expansion $R_\infty = \sum_{k \geq 0} \sigma_k$ with $\sigma_k \leq S^k$, $S = \sum_{n \geq 2} u_n$. From it we derive a closed-form two-sided bracket $\log(1+S) \leq \delta \leq -\log(1-S)$, with S in closed digamma form (generic poles) and confluent polygamma form (degenerate poles; e.g. $S(1, 2, 1) = \pi^2/3 - 13/4$ exactly). Using the first three cluster sums $\sigma_1, \sigma_2, \sigma_3$ in exact closed form (machine-verified against brute-force enumeration to $\sim 10^{-54}$) to seed an exact downward recurrence, we evaluate δ to certified high precision: $\delta(1, 0, 1)$ to ≥ 44 digits, with a nine-triple table, each value certified two internal ways and cross-checked by a fully independent forward/Neville extrapolation. An integer-relation (PSLQ/`identify`) search returns no low-height closed form, and a precision-stability test shows the uncontrolled relations to be numerical artifacts; we record δ as conjecturally a new transcendental constant of the family. Finally, the finitary structural core — the Casoratian identity, the monotone-bounded convergence $R_n \rightarrow R_\infty$, the closed-form bracket $1+S \leq R_\infty \leq 1/(1-S)$, the independence-polynomial identity $R_n = \sum_T \prod_{i \in T} u_i$, and the finite-window cluster core $\sigma_2 = e_2 - a_1$ — is machine-checked in Lean 4/Mathlib with clean axiom cones (no `sorry`). Version 1.3 enlarges this machine-checked core (Section 8): the Casoratian is generalized to an arbitrary second-order linear recurrence over any commutative ring (the classical $(-1)^n$ being the unit-coefficient case), the independence identity is refined to its degree-graded form (a `PROVEN` order-3 cluster window object, though the σ_3 *closed form* stays structural), an m -term geometric-tail enclosure sharpening the bracket is proved, and an exact weighted partial-fraction telescoping identity is established for the $B = 0$ family.

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1 Introduction

The convergent denominators of the quadratic polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$, integers $A, C \geq 1$, $B \geq 0$, obey $q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$ with $b_k = Ak^2 + Bk + C$. A companion note [1] establishes the exact two-sided growth law

$$\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \frac{\kappa}{n} + O(1/n^2), \quad (1)$$

where r_1, r_2 are the roots of $Ax^2 + Bx + C$, $K_\Gamma = -\log(\Gamma(1 - r_1)\Gamma(1 - r_2))$ is the *product constant* (at $B = 0$ equal to $\log(\sinh(\pi\sqrt{C/A})/(\pi\sqrt{C/A}))$), $\kappa = \frac{1}{2}(B^2/A^2 + B/A - 2C/A)$, and

$$\delta = \log R_\infty > 0, \quad R_\infty = \lim_{n \rightarrow \infty} \frac{q_n}{\prod_{k=1}^n b_k} \quad (2)$$

is the *correction constant* produced by the $+q_{n-2}$ term of the recurrence. The product piece K_Γ is closed-form; the companion note claims *no* closed form for δ and treats it as a black box. This note opens the box.

Our contributions are, in increasing order of analytic depth:

- (i) **Structure.** R_∞ is exactly the independence polynomial of the path graph on $\{2, \dots, n\}$ in the limit, evaluated at $u_n = 1/(b_{n-1}b_n)$ (Proposition 1); equivalently a Mayer-type cluster expansion $R_\infty = \sum_k \sigma_k$ with $\sigma_k \leq S^k$.
- (ii) **A closed-form bracket.** $\log(1 + S) \leq \delta \leq -\log(1 - S)$ (Proposition 2), where $S = \sum_{n \geq 2} u_n$ is closed-form via a digamma residue sum, with a confluent polygamma form at degenerate poles; one degenerate value is exactly rational-plus- π^2 : $S(1, 2, 1) = \pi^2/3 - 13/4$.
- (iii) **Certified high precision.** Exact closed forms for $\sigma_1, \sigma_2, \sigma_3$ seed an exact downward recurrence, yielding δ to ≥ 44 digits for $(1, 0, 1)$ and a nine-triple table (Section 5); each value is certified by seed-order independence and self-convergence, and cross-checked by an independent extrapolation method.
- (iv) **Non-elementarity (conjectural).** An integer-relation search finds no low-height closed form, and a precision-stability test exhibits the uncontrolled relations as artifacts (Section 6).
- (v) **A machine-checked core.** The finitary structural facts on which (i)–(iii) rest — the Casoratian identity, the monotone-bounded convergence $R_n \rightarrow R_\infty$, and the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ (conditional on $\sum u_n < \infty$ and $S < 1$) — are formalized in Lean 4/Mathlib with clean axiom cones (Section 7). Section 8 adds v1.3 generalizations and refinements (a general Casoratian over any commutative ring, a degree-graded independence refinement, an m -term enclosure, and exact weighted telescoping for $B = 0$), keeping every analytic input an explicit hypothesis.

Relation to prior work. The denominators q_n satisfy the three-term recurrence $q_n = b_n q_{n-1} + q_{n-2}$ with polynomially growing coefficients $b_n = An^2 + Bn + C \rightarrow \infty$, a Poincaré-type difference equation. Such a recurrence has a two-dimensional solution space spanned by a *dominant* solution (here q_n) and a *recessive* (minimal) solution; convergence of the continued fraction to V is exactly the selection of the minimal ray by p_n/q_n , the content of Pincherle’s theorem (originally due to Pincherle; modern statement and attribution in [3], with minimal/recessive solutions in [7, §3.6]). The leading large- n asymptotics of growing-coefficient recurrences are governed by Perron’s theorem [4] and the

Birkhoff–Trjitzinsky/Wong–Li asymptotic theory [5, 6]; applied to q_n these produce precisely the dominant terms $n \log A + 2 \log(n!) + \frac{B}{A} \log n$ and the Gamma-product constant K_Γ of the companion law (1). What that theory leaves undetermined is the additive *constant* — the amplitude relating the recessive solution to the product scale $\prod_k b_k$. The present note isolates and characterizes that constant, $\delta = \log R_\infty$: we show R_∞ is the path independence polynomial at the weights $u_n = 1/(b_{n-1}b_n)$ (Proposition 1), from which the closed-form bracket and the certified values below follow. To our knowledge neither this combinatorial identity for the amplitude nor the digamma closed form for its bracketing series S appears in that literature.

Standing assumptions. Throughout A, C are integers ≥ 1 and B is an integer ≥ 0 , so $b_k = Ak^2 + Bk + C \geq Ak^2 + C \geq A + C \geq 2 > 0$ for all $k \geq 1$, $q_n > 0$, and $P_n := \prod_{k=1}^n b_k > 0$. The roots of $Ax^2 + Bx + C$ satisfy $r_1 + r_2 = -B/A \leq 0$ and $r_1 r_2 = C/A > 0$, hence form either a complex-conjugate pair (purely imaginary exactly when $B = 0$) or two negative reals; in all cases $\operatorname{Re} r_i \leq 0$.

Epistemic convention. We use the four-class SIARC partition. **PROVEN** = machine-checked in Lean with a clean axiom cone (a subset of `{propxt, Classical.choice, Quot.sound}` and no `sorryAx`); **STRUCTURAL** = complete elementary hand proof, not yet formalized; **VERIFIED** = confirmed numerically for the stated finite cases; **CONJECTURED** = supported by evidence, not established. Each numbered result below is tagged accordingly, and Section 9 collects the grading. All numerical claims are reproduced by the scripts named in Section 10.

2 The scaled recurrence and the Casoratian frame

Let $R_n := q_n/P_n$, so $R_0 = R_1 = 1$. The convergent numerators p_n ($p_{-1} = 1, p_0 = 1$) satisfy the same second-order recurrence; set $P_n^{\text{num}} := p_n/P_n$.

Lemma 1 (Scaled recurrence; STRUCTURAL, Lean core in Section 7). *For all $n \geq 2$,*

$$R_n = R_{n-1} + u_n R_{n-2}, \quad u_n := \frac{1}{b_{n-1}b_n} > 0.$$

Consequently (R_n) is strictly increasing, bounded above by $\prod_{k \geq 2} (1 + u_k) < \infty$, and converges to a limit $R_\infty \in (1, \infty)$ with $R_\infty - R_n = O(1/n^3)$; thus $\delta = \log R_\infty > 0$.

Proof. Divide $q_n = b_n q_{n-1} + q_{n-2}$ by $P_n = b_n P_{n-1} = b_n b_{n-1} P_{n-2}$ to get $R_n = R_{n-1} + R_{n-2}/(b_{n-1}b_n)$. Since $u_n > 0$ and $R_0, R_1 > 0$, induction gives $R_n > 0$ and the increment $R_n - R_{n-1} = u_n R_{n-2} > 0$, so (R_n) strictly increases; monotonicity gives $R_{n-2} \leq R_{n-1}$, whence $R_n \leq R_{n-1}(1 + u_n)$ and $R_n \leq \prod_{k=2}^n (1 + u_k)$. As $b_{k-1}b_k \geq A^2(k-1)^2 k^2$, the series $\sum_k u_k = O(\sum k^{-4})$ converges, so the product converges and (R_n) is bounded; a monotone bounded sequence converges to $R_\infty \geq R_2 = 1 + u_2 > 1$. The tail estimate $R_\infty - R_n = \sum_{m>n} (R_m - R_{m-1}) = \sum_{m>n} u_m R_{m-2} \leq R_\infty \sum_{m>n} u_m = O(1/n^3)$, where the first equality telescopes the increment $R_m - R_{m-1} = u_m R_{m-2}$, follows by comparison with $\int_n^\infty t^{-4} dt$ since $u_m \sim (Am^2)^{-2}$. $\square$

Lemma 2 (Casoratian; PROVEN, Section 7). *The discrete Wronskian of p_n, q_n is $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$; hence, in scaled variables,*

$$P_n^{\text{num}} R_{n-1} - P_{n-1}^{\text{num}} R_n = \frac{(-1)^{n-1}}{b_n \left(\prod_{k=1}^{n-1} b_k \right)^2}.$$

The initial data $p_{-1}=1, p_0=1, q_{-1}=0, q_0=1$ give $p_1=b_1+1$ and $q_1=b_1$, so Lemma 2 is exactly the convergent Casoratian $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$ of Theorem 1 re-indexed by $n \mapsto n+1$; the Lean statement uses that shifted index and the equivalent sign $(-1)^n$.

The Wronskian decays like $P_n^{-2} \asymp (n!)^{-4}$, which is why the *value* $V = \lim p_n/q_n$ converges super-geometrically (matched to 77–117 digits across the test triples already at $N \approx 4000$, and to the 129 digits used in Section 6 at larger N) while $R_\infty - R_n$ decays only like $1/n^3$. Consistently, the $\kappa/n + O(1/n^2)$ remainder in (1) is the Euler–Maclaurin tail of $\log P_n = \sum_{k \leq n} \log b_k$, whereas $\log(q_n/P_n) = \delta + O(1/n^3)$ contributes only to the constant $K_\Gamma + \delta$ and a strictly smaller remainder absorbed into $O(1/n^2)$. The slow convergence of R_n is what forces the dedicated high-precision method of Section 5; a naive forward recurrence stalls at ~ 25 –38 digits.

3 R_∞ as an independence polynomial

The recurrence $R_n = R_{n-1} + u_n R_{n-2}$ is *not* the classical continuant recurrence $K_n = x_n K_{n-1} + K_{n-2}$: the weight u_n multiplies the *second-order* term. It is instead the transfer recurrence of an independence polynomial.

Proposition 1 (Independence-polynomial identity; PROVEN (Lean, Theorem 1)). *For every n ,*

$$R_n = \sum_{\substack{T \subseteq \{2, \dots, n\} \\ \text{no two consecutive}}} \prod_{i \in T} u_i,$$

the independence polynomial of the path graph on the vertices $\{2, \dots, n\}$ (edges between consecutive integers), evaluated at vertex weights u_i . Letting $n \rightarrow \infty$,

$$R_\infty = \sum_{k \geq 0} \sigma_k, \quad \sigma_0 = 1, \quad \sigma_1 = S = \sum_{n \geq 2} u_n, \quad \sigma_k = \sum_{\substack{T \subseteq \{2, 3, \dots\} \\ |T|=k \\ \text{gaps} \geq 2}} \prod_{i \in T} u_i. \quad (3)$$

Because each non-consecutive k -subset is one of the unrestricted k -tuples (which double-count order, repeats, and adjacency), $0 \leq \sigma_k \leq S^k$, so the series converges geometrically.

Proof. Both sides satisfy the same recurrence and initial data. The empty set gives the constant term 1 ($R_1 = 1$); a non-consecutive subset $T \subseteq \{2, \dots, n\}$ either omits n (contributing to R_{n-1}) or contains n , in which case it omits $n-1$ and the rest is a non-consecutive subset of $\{2, \dots, n-2\}$ with the factor u_n (contributing $u_n R_{n-2}$). This is exactly $R_n = R_{n-1} + u_n R_{n-2}$, and the $n = 1, 2$ base cases match. The bound $\sigma_k \leq S^k$ is the stated over-count. $\square$

The cluster identity (3) is verified *exactly* (subset dynamic-programming equals R_n for all tested triples and all n) by the script `continuant_verify.py`.

4 A closed-form two-sided bracket

Proposition 2 (Bracket; PROVEN (Lean, conditional core; Section 7), VERIFIED all triples). *With $S = \sum_{n \geq 2} u_n < 1$,*

$$1 + S \leq R_\infty \leq \frac{1}{1 - S} \quad \Longleftrightarrow \quad \log(1 + S) \leq \delta \leq -\log(1 - S).$$

Proof. From $R_\infty = 1 + \sum_{n \geq 2} u_n R_{n-2}$ and $R_{n-2} \geq 1$ comes the lower bound $R_\infty \geq 1 + S$. For the upper bound, by (3) and $\sigma_k \leq S^k$, $R_\infty = \sum_k \sigma_k \leq \sum_k S^k = 1/(1 - S)$ (using $S < 1$: each $b_k = Ak^2 + Bk + C$ is increasing in each of A, B, C for $k \geq 1$, so $u_n = 1/(b_{n-1}b_n)$ and hence $S = \sum_{n \geq 2} u_n$ are maximized at the minimal admissible triple $(1, 0, 1)$, giving $S \leq S(1, 0, 1) \approx 0.1307 < 1$). Apply log. $\square$

The bracket is deliberately coarse: the lower bound keeps only $\sigma_0 + \sigma_1$ and the upper bound replaces each σ_k by its geometric majorant S^k . The exact cluster forms σ_2, σ_3 of Section 5 (Appendix A) sharpen it well past the bracket.

Remark 1 (Machine-checked, cluster-free route). *The R_∞ -form of the bracket is formalized in Lean as `rinf_bracket` (Section 7), conditional only on the elementary analytic input that the partial sums $S_n = \sum_{k=2}^n u_k$ are bounded above and $S = \sup_n S_n < 1$. The formalization does not use the cluster expansion (3): it rests on the two finitary inequalities $1 + S_n \leq R_n$ (`rseq_ge_one_add_psum`) and $(\prod_{k=2}^n (1 + u_k))(1 - S_n) \leq 1$ (`rmaj_mul_one_sub_psum_le_one`), the latter from the exact identity $(1 - S_n) - (1 + u)(1 - S_n - u) = u S_n + u^2 \geq 0$. The δ -form follows by monotonicity of log.*

It remains to put S in closed form. Write $b_n = A(n - r_1)(n - r_2)$, so $1/(b_{n-1}b_n)$ has its poles in the multiset $\{r_1, r_2, 1 + r_1, 1 + r_2\}$.

Proposition 3 (Closed form for S ; VERIFIED to 35–62 digits). *If the four poles are distinct,*

$$S = - \sum_{\rho} \text{Res}_{\rho} \psi(2 - \rho), \quad \text{Res}_{\rho} = \frac{1}{A^2 \prod_{\sigma \neq \rho} (\rho - \sigma)}, \quad (4)$$

with ψ the digamma function (the sum of residues vanishes, $\sum_{\rho} \text{Res}_{\rho} = 0$, reflecting $1/(b_{n-1}b_n) = O(n^{-4})$, so the individually divergent ψ -pieces combine convergently). For repeated poles, partial fractions over the multiset of multiplicities m_{ρ} give the confluent form

$$S = - \sum_{\rho} a_{\rho,1} \psi(2 - \rho) + \sum_{\rho} \sum_{j \geq 2} a_{\rho,j} \frac{(-1)^j \psi^{(j-1)}(2 - \rho)}{(j - 1)!}, \quad (5)$$

$a_{\rho,j}$ the Laurent coefficients of $1/(b_{n-1}b_n)$ at ρ , with $\sum_{\rho} a_{\rho,1} = 0$.

Equations (4)–(5) are the termwise sums $\sum_{n \geq 2} (n - \rho)^{-1}$ (regularized, $j = 1$) and $\sum_{n \geq 2} (n - \rho)^{-j} = (-1)^j \psi^{(j-1)}(2 - \rho)/(j - 1)!$ ($j \geq 2$), standard polygamma series [7, §5.15]. Both are confirmed against direct summation (`mpmath.nsum`) for distinct- and repeated-pole triples (`S_closed_form.py`, `S_confluent.py`). A clean degenerate value: for $(A, B, C) = (1, 2, 1)$, $b_n = (n + 1)^2$ has poles $\{-1, -1, 0, 0\}$ and

$$S(1, 2, 1) = \frac{\pi^2}{3} - \frac{13}{4} = 0.0398681336964528729 \dots, \quad (6)$$

matched to 10^{-40} . (Under the standing assumptions the poles of $1/(b_{n-1}b_n)$, namely $r_1, r_2, 1 + r_1, 1 + r_2$, each have real part $\leq 1 < 2$ because $\text{Re } r_i \leq 0$; hence no summation index $n \geq 2$ is a pole and the residue and confluent forms apply throughout.) Table 1 shows the bracket on representative triples.

Table 1: The closed-form bracket $\log(1 + S) \leq \delta \leq -\log(1 - S)$ on representative triples; δ from the high-precision method of Section 5.

(A, B, C)	S	$\log(1 + S)$	δ	$-\log(1 - S)$
(1, 0, 1)	0.13066961899	0.12281004	0.12385719436	0.14003204
(1, 0, 2)	0.080228466794	0.077172562	0.077741108481	0.083629973
(2, 0, 1)	0.045720089039	0.044705729	0.044814419813	0.046798243
(1, 1, 1)	0.065740306948	0.063669682	0.064033816183	0.068000835
(3, 2, 5)	0.0068561031151	0.0068327069	0.0068373312816	0.0068797142

5 Certified high-precision values

Because $R_n \rightarrow R_\infty$ only like $1/n^3$, the forward recurrence is useless beyond a few dozen digits. We instead run the *exact downward recurrence*

$$T(m) = T(m+1) + u_m T(m+2), \quad T(\infty) = 1, \quad R_\infty = T(2), \quad (7)$$

where $T(m)$ is the independence polynomial of the tail $\{m, m+1, \dots\}$ (so $T(m) = \sum_{k \geq 0} \sigma_k(m)$ with $\sigma_k(m)$ the order- k cluster sum starting at m). The recurrence (7) is exact, so *all* error lives in the seed $T(M+1), T(M+2)$ at the truncation index M . Truncating the cluster series at order p leaves seed error $\sim \sigma_{p+1} = O(M^{-3(p+1)})$, i.e. about $3(p+1)$ correct digits per decade of M . We use $p = 3$ (~ 12 digits/decade), for which exact closed forms are available:

$$\sigma_2 = \frac{1}{2}(p_1^2 - p_2) - a_1, \quad (8)$$

$$\sigma_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3) - a_1p_1 + c + t, \quad (9)$$

with power sums $p_j = \sum_{n \geq m} u_n^j$ (and $p_1 = \sigma_1(m) = S(m)$ from Proposition 3, exact via ψ), and adjacency sums $a_1 = \sum_n u_n u_{n+1}$, $c = \sum_n u_n u_{n+1}(u_n + u_{n+1})$, $t = \sum_n u_n u_{n+1} u_{n+2}$.

Proposition 4 (Cluster closed forms; σ_2 core PROVEN (Lean, Theorem 1), σ_3 STRUCTURAL (Appendix A), VERIFIED to $\sim 10^{-54}$). *Equations (8)–(9) hold: the unrestricted elementary symmetric sums $e_2 = \frac{1}{2}(p_1^2 - p_2)$, $e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3)$ corrected for adjacency by inclusion–exclusion (Appendix A). They agree with the brute-force enumeration of non-consecutive 2- and 3-subsets for (A, B, C) equal to (1, 0, 1), (1, 1, 1), (2, 1, 3), (1, 2, 1) to better than 10^{-40} (absolute residuals $\sim 10^{-54}$ – 10^{-55}).*

This is the gate of `verify_clusters.py`: no derived combinatorial identity is trusted until machine-checked against brute force. With the gate passed, the downward recurrence (7) seeded by $1 + \sigma_1(M) + \sigma_2(M) + \sigma_3(M)$ gives the values in Table 2.

For the reference triple (1, 0, 1) we certify δ *three* independent ways at $M = 10^6$ (working precision 170 digits): seed-order independence (order-2 vs order-3 seed agree to 56 digits; order-3 vs order-3-plus-leading- σ_4 to 75 digits), self-convergence ($M = 10^5$ vs 10^6 agree to 47 digits), and a fully independent forward continuant with Neville extrapolation in $1/N$ (agrees to ~ 25 digits, its own resolution ceiling). The conservative reliable count is ≥ 44 digits:

$$\delta(1, 0, 1) = 0.12385719436062639272850498970259084096757955 \dots \quad (10)$$

Table 2: Certified high-precision $\delta(A, B, C) = \log R_\infty$ via the exact order-3 cluster seed and downward recurrence (7) at $M = 10^5$ (working precision 90 digits), each value self-checked against $M = 2.5 \times 10^4$ and cross-checked by independent forward/Neville extrapolation. “rel. dig.” is the conservatively certified number of correct digits.

(A, B, C)	$\delta = \log R_\infty$ (first 30 digits)	rel. dig.
(1, 0, 1)	0.123857194360626392728504989702	44 [†]
(1, 0, 2)	0.077741108480647487187139168954	39
(1, 0, 3)	0.055023822866781361284024549668	39
(2, 0, 1)	0.044814419812755566106936562111	40
(3, 0, 1)	0.023024249675539505266856933463	41
(1, 1, 1)	0.064033816182909523022426151229	39
(1, 2, 1)	0.039254537227423694454156302525	39
(2, 1, 3)	0.017942479040119357169479600823	40
(3, 2, 5)	0.006837331281558730523182815470	41

[†] certified to ≥ 44 digits at $M = 10^6$, dps= 170; see (10).

Two robustness notes. The degenerate triple (1, 2, 1) has coincident poles, so the simple-pole residue form (4) is singular there; the seed’s $\sigma_1(m) = S(m)$ must use the exact confluent polygamma form (5) (a direct numerical tail sum loses precision at large m). With that substitution (1, 2, 1) matches the others (39 digits), and its $S(2)$ reproduces (6) exactly. Separately, the independent forward/Neville cross-check must start the scaled continuant at the index that includes the u_2 term, or it converges to a different limit; with the correct start it agrees with the cluster value to ~ 25 digits across all tested triples.

On the previously reported “80-digit” figure. An earlier internal record cited an 80-digit value for $\delta(1, 0, 1)$. The present method reaches 80 digits in principle at $M = 10^7$ (consistent with the ~ 12 digit/decade rate), but we *certify* only the ≥ 44 digits of (10) at feasible cost; nothing is asserted beyond the certified count.

6 No low-height closed form for δ

Proposition 5 (Non-elementarity; CONJECTURED, rigorous null). *No low-height integer relation is found for V , R_∞ , or δ against natural constant bases. Specifically: $\text{identify}(V) = \text{None}$ at 100 digits (the strongest null, V being the natural “answer”, known to 129 digits via the fast p_n/q_n ratio); $\text{identify}(\delta) = \text{None}$; PSLQ over $\{1, \pi, \log 2, \gamma, \text{Catalan}, \zeta(3)\}$ with $\max |c_i| \leq 200$ returns no relation; and R_∞ — the natural algebraic candidate, tested by PSLQ on $\{1, R_\infty, R_\infty^2, R_\infty^3, R_\infty^4\}$ — yields no integer relation of degree ≤ 4 and height $\leq 10^4$ (δ and V receive the *identify* null above).*

Uncontrolled PSLQ does return high-height vectors, but `pslq_stability.py` shows them to be artifacts: the recovered vector wanders with working precision ($[30, 110, \dots] \rightarrow [-244, \dots] \rightarrow [1330, \dots] \rightarrow [-25296, \dots]$) and the residual floors at $\sim 10^{-\text{dps}}$ rather than shrinking toward true zero — whereas a genuine integer relation is precision-invariant. We therefore record δ as conjecturally a genuine new transcendental constant of the quadratic family, not reducible to the product constant K_Γ . Irrationality of δ is open. A null integer-relation result is, we stress, a complete and honest answer, not a gap.

7 The machine-checked finitary core

The finitary facts underlying Sections 2–4 are formalized in Lean 4 with Mathlib (toolchain pinned: lean4:v4.30.0, Mathlib rev=v4.30.0). Each declaration was checked with `#print axioms`; every axiom cone is a subset of the Mathlib defaults `{propext, Classical.choice, Quot.sound}` with no `sorryAx`, the source contains no `sorry/admit`, and the build is warning- and error-free (exit 0).

Theorem 1 (Formalized core; PROVEN). *The following are machine-checked with clean axiom cones:*

- *casoratian_step, casoratian_eq, pq_casoratian*: the Casoratian of two solutions of $s_{n+2} = b_{n+2}s_{n+1} + s_n$ flips sign each step, hence equals $(-1)^n$ times its initial value; for the convergents $(p_0=1, p_1=b_1+1, q_0=1, q_1=b_1)$ this is $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$ (over an arbitrary *CommRing*).
- *rseq_ge_one, rseq_mono, rseq_monotone, rmaj_ge_one, rmaj_mono, rseq_le_rmaj*: the scaled sequence $r_{n+2} = r_{n+1} + u_{n+2}r_n$ ($r_0=r_1=1$, $u \geq 0$) satisfies $r_n \geq 1$, is monotone, and is dominated by the partial products $\prod_{k=2}^n (1 + u_k)$ (over $\mathbb{R}$).
- *rseq_bddAbove_of_rmaj, rseq_tendsto_ciSup, rseq_tendsto_of_rmaj_bddAbove*: monotone convergence — if the partial products are bounded above (equivalently $\sum u_n < \infty$) then $R_n \rightarrow \sup_n R_n$, i.e. R_∞ exists.
- *psum, psum_nonneg, rseq_ge_one_add_psum, rmaj_mul_one_sub_psum_le_one, rinf_bracket, rseq_tendsto_under_psum_bdd*: the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ (Proposition 2), conditional on the partial sums $S_n = \sum_{k=2}^n u_k$ being bounded above and $S = \sup_n S_n < 1$; the finitary core is $1 + S_n \leq R_n$ and $(\prod_{k=2}^n (1 + u_k))(1 - S_n) \leq 1$ (over $\mathbb{R}$).
- *NoTwoConsec.erase, indepPoly_zero, indepPoly_one, filter_not_mem_top, sum_filter_mem_top, indepPoly_succ_succ, indepPoly_eq_rseq*: the independence-polynomial identity (Proposition 1), unconditionally. The weighted independence polynomial of the path graph on $\{2, \dots, n\}$, $\text{indepPoly}(u, n) = \sum_T \prod_{i \in T} u_i$ over no-two-consecutive $T \subseteq \{2, \dots, n\}$, equals R_n ; proved by the deletion–contraction split at the top vertex (over $\mathbb{R}$, no positivity needed).
- *sigma2win_eq*: the finite-window inclusion–exclusion core of the σ_2 closed form (Proposition 4, Appendix A), unconditionally: over any window $[m, M]$, the no-two-consecutive pair sum equals $\sigma_2 = e_2 - a_1$ (every ordered pair is consecutive or not). The Newton reduction $e_2 = \frac{1}{2}(p_1^2 - p_2)$ and the σ_3 closed form are not yet formalized (see STRUCTURAL below).

This is the *conditional-core* pattern: the finitary algebra and order structure are proved unconditionally, while the single analytic input — that the partial products are bounded above — is taken as an explicit labelled hypothesis (it is the convergence of $\sum u_n = O(\sum k^{-4})$, elementary but analytic). Out of scope for the formalization, and proved by hand/numerics above, are: the *value* of R_∞ , the closed forms for S , the product constant K_Γ and the Stirling/Gamma asymptotics of (1), and the non-elementarity conjecture.

8 Formalized generalizations and refinements

Version 1.3 extends the machine-checked core of Section 7 in four independent directions, keeping the same conditional-core discipline: all combinatorial and algebraic content is unconditional, and each analytic input is an explicit labelled hypothesis. Every declaration named below was checked with `#print axioms` to a cone within `{propext, Classical.choice, Quot.sound}` (`IsSol.isSolG`

depends on *no* axioms at all), with no `sorryAx`, no `sorry/admit` in source, and build exit 0; the harness `Check.lean` now reports 58 such cones.

Theorem 2 (v1.3 formal additions; PROVEN). *The following hold with clean axiom cones.*

- (A) **General Casoratian over a commutative ring.** `IsSol.isSolG`, `casoratian_stepG`, `casoratian_eqG`, `pq_casoratianG`: over any `CommRing` R , for arbitrary coefficient sequences $c, d : \mathbb{N} \rightarrow R$, every solution of the general second-order linear recurrence $s_{n+2} = c_{n+2} s_{n+1} + d_n s_n$ has Casoratian $W_n = p_n q_{n-1} - p_{n-1} q_n$ obeying $W_{n+1} = -d_n W_n$, hence $W_n = (\prod_{k < n} (-d_k)) W_0$ and $p_{n+1} q_n - p_n q_{n+1} = (\prod_{k < n} (-d_k)) (p_1 q_0 - p_0 q_1)$. `casoratian_eqG.recovers_classical` specializes this at $d \equiv 1$ to the $(-1)^n$ identity of Lemma 2: the v1.2 Casoratian results are exactly its unit-coefficient case (a faithfulness witness for the generalization).
- (B) **Degree-graded independence and the σ_3 window object.** `indepKWin`, `indepKWin_succ_succ`, `indepPolyWin_eq_sum_indepKWin`, `sigma3Win_eq`, `sigma3Win_succ_succ`: the independence polynomial of Proposition 1 is split into its degree- k graded pieces $\text{indepKWin}(u, k, a, b) = \sum_{|T|=k} \prod_{i \in T} u_i$ over no-two-consecutive $T \subseteq [a, b]$, each satisfying the deletion-contraction recurrence; summing over k returns `indepPolyWin`. The order-3 piece is the cluster window object $\text{sigma3Win} = \text{indepKWin}(u, 3, \cdot, \cdot)$, whose recurrence is proved. The σ_3 closed form (9) is deliberately **not** promoted: only the graded combinatorial object and its recurrence are PROVEN; the identity $\sigma_3 = \frac{1}{6}(p_1^3 - 3p_1 p_2 + 2p_3) - a_1 p_1 + c + t$ remains STRUCTURAL (Appendix A) and VERIFIED ($\sim 10^{-54}$), because the $a_1 p_1$ cross-term regroups monomials across the index cube and is not a termwise identity.
- (C) **An m -term enclosure refining the bracket.** `tsum_geomTail`, `Rinf_enclosure`, `delta_enclosure`, `enclosure_nonvacuous`, `m1_recovers_S_bracket`, `m1_delta_eq_S_bracket`, `H2_zero_forces_sigma0_le_one`: with $L_m = \sum_{k=0}^m \sigma_k$, the partial-sum/geometric-tail sandwich gives $L_m \leq R_\infty \leq L_m + S^{m+1}/(1 - S)$ and, when $L_m > 0$, $\log L_m \leq \delta \leq \log(L_m + S^{m+1}/(1 - S))$, conditional on the two labelled hypotheses (H1) $R_\infty = \sum_k \sigma_k$ and (H2) $0 \leq \sigma_k \leq S^k$. At $m = 1$ both endpoints reduce exactly to the bracket of Proposition 2, and (H2) at $k = 0$ forces $\sigma_0 \leq 1$ — the two points raised in review, now proved rather than observed.
- (D) **Exact weighted telescoping for the $B = 0$ family.** `partialFraction`, `sum_Ico_telescope`, `weighted_telescope`, `weighted_telescope_value`, `weighted_sum_tendsto`: over any field, with $\beta_m = A m^2 + C$ (the $B = 0$ denominator) nonzero on the summation window, the weight $A(2m + 1) = \beta_{m+1} - \beta_m$ makes $A(2m + 1)/(\beta_m \beta_{m+1})$ telescope to the exact finite value

$$\sum_{m=1}^{N-1} \frac{A(2m+1)}{\beta_m \beta_{m+1}} = \frac{1}{A+C} - \frac{1}{AN^2+C},$$

with no characteristic restriction (the 2 only multiplies, never divides). Over $\mathbb{R}$ the infinite sum equals $1/(A+C)$, PROVEN conditional on the single labelled analytic input $1/(AN^2+C) \rightarrow 0$. The unweighted truncation $\sum 1/(\beta_{m-1} \beta_m)$ has, by contrast, no rational closed form in N (Gosper's algorithm returns no telescoper); its coth/digamma limit needs real special functions and is left out of scope, numerically VERIFIED only.

No numerically VERIFIED statement is silently upgraded by these additions — in particular the σ_3 closed form and the unweighted telescoping limit remain non-PROVEN — and the epistemic

ledger `claims.jsonl` records, per result, the grade and the hypotheses under which each conditional statement may be cited downstream.

9 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 1): the Casoratian identity, the existence of $R_\infty = \lim R_n$ via monotone–bounded convergence, the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ (Proposition 2), the independence-polynomial identity (Proposition 1, unconditional), and the finite-window inclusion–exclusion core $\sigma_2 = e_2 - a_1$ of the σ_2 closed form (Proposition 4, unconditional); the bracket and limit are conditional on the elementary analytic input $\sum u_n < \infty$ (and $S < 1$ for the bracket), the independence-polynomial and σ_2 cores require no analytic input. Version 1.3 (Theorem 2) further proves: the general Casoratian over any commutative ring (with the $(-1)^n$ case recovered as a faithfulness witness), the degree-graded independence refinement and the order-3 cluster window object `sigma3Win` (the σ_3 *closed form* remains structural/verified and is *not* promoted), the m -term enclosure $L_m \leq R_\infty \leq L_m + S^{m+1}/(1 - S)$ (conditional on $R_\infty = \sum_k \sigma_k$ and $0 \leq \sigma_k \leq S^k$), and the exact weighted telescoping identity for the $B = 0$ family (its conditional limit resting only on $1/(AN^2 + C) \rightarrow 0$).
- **STRUCTURAL** (complete elementary hand proof; not yet formalized): the scaled recurrence and tail estimate (Lemma 1), the closed forms for S (Proposition 3), and — completing the cluster forms beyond the Lean σ_2 core above — the Newton reduction $e_2 = \frac{1}{2}(p_1^2 - p_2)$, $e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3)$ and the σ_3 closed form (Proposition 4; the inclusion–exclusion derivation of σ_2, σ_3 is Appendix A).
- **VERIFIED** (numerically): the cluster identity exactly (`continuant_verify.py`); σ_2, σ_3 and each step of their inclusion–exclusion derivation to $\sim 10^{-54}$ (`verify_clusters.py`, `verify_inclusion_exclusion.py`); S closed forms to 35–62 digits vs direct summation; the bracket on all tabulated triples; and the δ values of Table 2, certified ≥ 39 digits each and ≥ 44 for $(1, 0, 1)$, with an independent cross-check.
- **CONJECTURED**: δ has no low-height elementary closed form (Proposition 5); δ is a new transcendental constant of the family. Irrationality of δ is open.

10 Code and data availability

Every numerical claim is reproduced by the following scripts (mpmath): the harness `harness_rinf.py` (two-solution frame, R_∞ , value V); `continuant_verify.py` (exact independence-polynomial check); `S_closed_form.py`, `S_confluent.py` (closed forms for S); `verify_clusters.py` and `verify_inclusion_exclusion.py` (the σ_2, σ_3 gate and the step-by-step inclusion–exclusion check, Proposition 4, Appendix A); `verify_sigma3_finite_window.py` (exact-rational gate for the order-3 window object) and `verify_telescope_b0.py` (exact-rational gate for the weighted telescoping identity), both Theorem 2; `hp_delta_v2.py` and `finalize_m2.py` (the high-precision δ values and Table 2, output `delta_values_m2.txt`); `verify_forward_crosscheck.py` (the independent cross-check); `hp_probe.py`, `pslq_stability.py` (the integer-relation nulls, Proposition 5); and the Lean package `lean/pcf_continuant` (`Basic.lean`, `Bracket.lean`, `Indep.lean`, `Sigma.lean`, `Topic3.lean`, `Topic5.lean`, `Check.lean`; Theorems 1 and 2). Version 1.0 of this note is archived at Zenodo (version DOI 10.5281/zenodo.20578401, concept DOI 10.5281/zenodo.20578400) as a

supplement to the companion growth-law note [1]; versions 1.1–1.2 added the machine-checked closed-form bracket (`rinf_bracket`), the independence-polynomial identity (`indepPoly_eq_rseq`), and the finite-window σ_2 inclusion–exclusion core (`sigma2win_eq`); version 1.3 adds the generalizations and refinements of Theorem 2 (general Casoratian, degree-graded independence and the order-3 window object, the m -term enclosure, and exact weighted telescoping for $B = 0$). The harness `Check.lean` now reports 58 axiom cones, each a subset of `{propext, Classical.choice, Quot.sound}` with no `sorryAx`. The Lean sources and the verification scripts named above accompany the deposit in its public repository, <https://github.com/papanokechi/pcf-delta>.

A Inclusion–exclusion derivation of the cluster closed forms

This appendix proves Equations (8)–(9) by elementary inclusion–exclusion, upgrading Proposition 4 from a numerically VERIFIED identity to a complete STRUCTURAL hand proof; the brute-force gate of `verify_clusters.py` and the step-by-step gate of `verify_inclusion_exclusion.py` then confirm every line to $\sim 10^{-54}$.

Fix the tail $\{m, m+1, \dots\}$ with vertex weights $u_i > 0$ and power sums $p_j = \sum_{i \geq m} u_i^j$. Recall from (3) that $\sigma_k = \sum \prod_{i \in T} u_i$ runs over k -subsets T with no two consecutive indices, and let $e_k = \sum_{|T|=k} \prod_{i \in T} u_i$ be the *unrestricted* elementary symmetric sum. Newton’s identities give

$$e_2 = \frac{1}{2}(p_1^2 - p_2), \quad e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3). \quad (11)$$

Derivation of (8). Each unrestricted pair $\{i < j\}$ is either non-consecutive (counted by σ_2) or consecutive ($j = i+1$); the consecutive pairs sum to $a_1 = \sum_i u_i u_{i+1}$. Hence $\sigma_2 = e_2 - a_1 = \frac{1}{2}(p_1^2 - p_2) - a_1$. $\square$

Derivation of (9). Classify the unrestricted triples $\{i < j < k\}$ by their number of consecutive pairs. Let $\mathcal{B}$ be those with at least one, so $\sigma_3 = e_3 - \Sigma(\mathcal{B})$, where Σ denotes the weighted sum $\sum \prod_{i \in T} u_i$. A triple has either *exactly one* consecutive pair — an adjacent pair plus an isolated third vertex; call this set $\mathcal{B}_1$ — or *two* — a run $\{i, i+1, i+2\}$; call this $\mathcal{B}_2$ — and $\Sigma(\mathcal{B}_2) = \sum_i u_i u_{i+1} u_{i+2} = t$.

Expand the product of the adjacency sum and the first power sum:

$$a_1 p_1 = \left(\sum_i u_i u_{i+1} \right) \left(\sum_r u_r \right) = \sum_i \sum_r u_i u_{i+1} u_r.$$

Split the inner sum according to whether $r \in \{i, i+1\}$. The terms with $r = i$ or $r = i+1$ sum to $\sum_i u_i u_{i+1} (u_i + u_{i+1}) = c$. The remaining terms, with $r \notin \{i, i+1\}$, each form a triple — the adjacent pair $\{i, i+1\}$ together with a distinct third vertex r — whose total we call W , so that $a_1 p_1 = c + W$. In W each member of $\mathcal{B}_1$ is counted once (its unique adjacent pair fixes $(i, i+1)$ and r is the isolated vertex), while each run $\{a, a+1, a+2\}$ in $\mathcal{B}_2$ is counted twice (as the pair $(a, a+1)$ with $r = a+2$, and as the pair $(a+1, a+2)$ with $r = a$). Thus $W = \Sigma(\mathcal{B}_1) + 2t$, and

$$\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t = (W - 2t) + t = W - t = a_1 p_1 - c - t.$$

Subtracting from e_3 in (11),

$$\sigma_3 = e_3 - \Sigma(\mathcal{B}) = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3) - a_1 p_1 + c + t. \quad \square$$

The script `verify_inclusion_exclusion.py` checks each labelled step independently against brute force over a finite window (e_2, e_3 against the elementary sums; a_1 against the adjacent pairs; $a_1 p_1 = c + W$; $W = \Sigma(\mathcal{B}_1) + 2t$; $\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t$; and the two closed forms), for $(A, B, C) \in \{(1, 0, 1), (1, 1, 1), (2, 1, 3), (1, 2, 1)\}$, all to $\sim 10^{-54}$.

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